Algorithm vs ML Algorithm

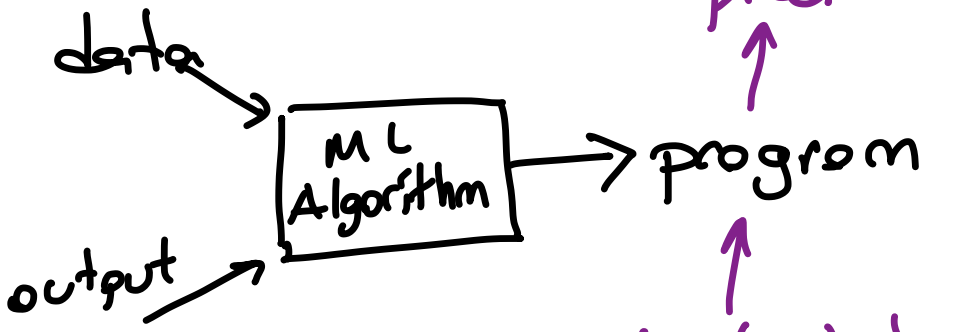
- sorting
- searching
- ...

① given a list of #s

task: to order these #s from smallest to largest

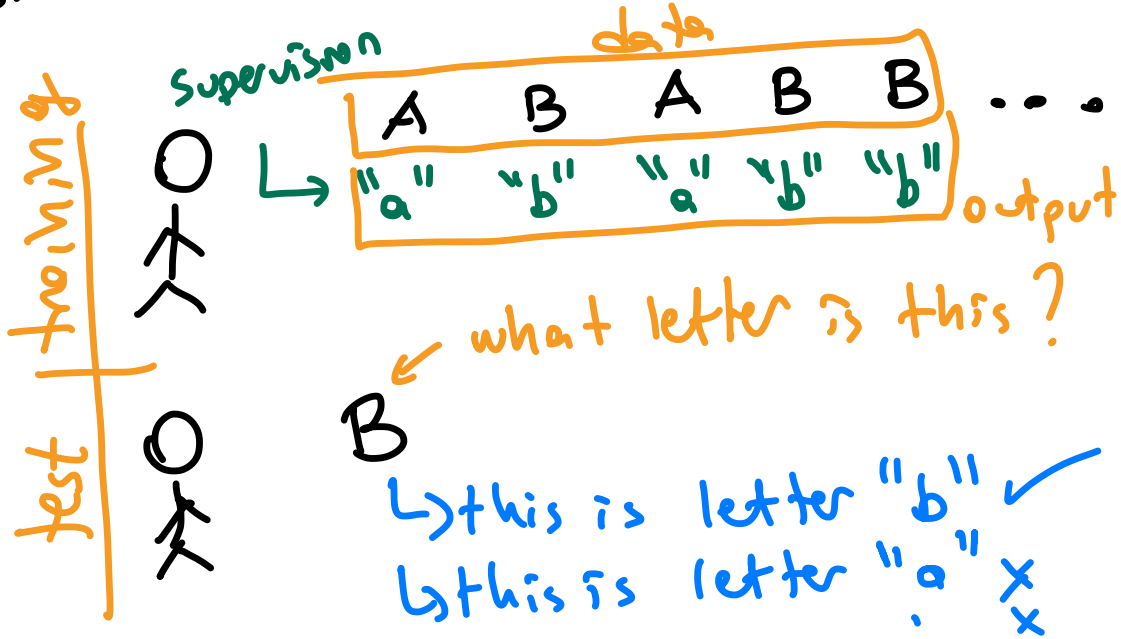


- the result is always correct
- deterministic
- fixed set of rules



⇒ subjectivity (example data, past data, historical data)

- unseen data
- out-of-sample data
- out-of-time data.



Machine Learning: programming computers to optimize learning a performance criterion using example data / past experience

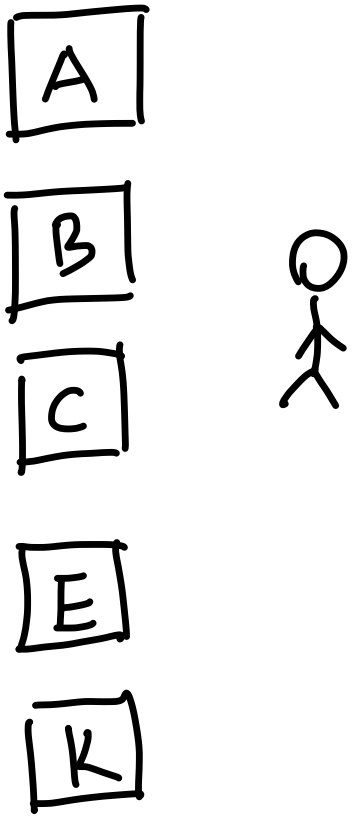
loss/error function

training data set

scaler ← X X X → matrices

↳ vectors

inputs



outputs

"a" ✓

"b" ✓

"c" ✓

"b" ✗

"n" ✗

	predictions	truth
Feb 18	8°C	12°C
Feb 19	6°C	8°C
Feb 20	4°C	2°C
Feb 21	2°C	4°C
Feb 22	8°C	6°C ✓
Feb 23	6°C	6°C
Feb 24	7°C	8°C

underestimation ↓

overestimation ↑

absolute error: $|y_i - \hat{y}_i|$

truth ← ↳ predicted

Supervised Learning:

training set $\mathcal{X} = \{(x_i, y_i)\}_{i=1}^{N}$

- N → # of data points
- $i = 1$ → index for data points
- x_i → ith data point
- y_i → ith label, ith target output

$$\mathcal{X} = \{(x_1, y_1), (x_2, y_2), \ldots\ldots, (x_N, y_N)\}$$

$$\mathcal{X} = \{([\,], \square)\}$$

scalers

↓ D-dimensional real vectors

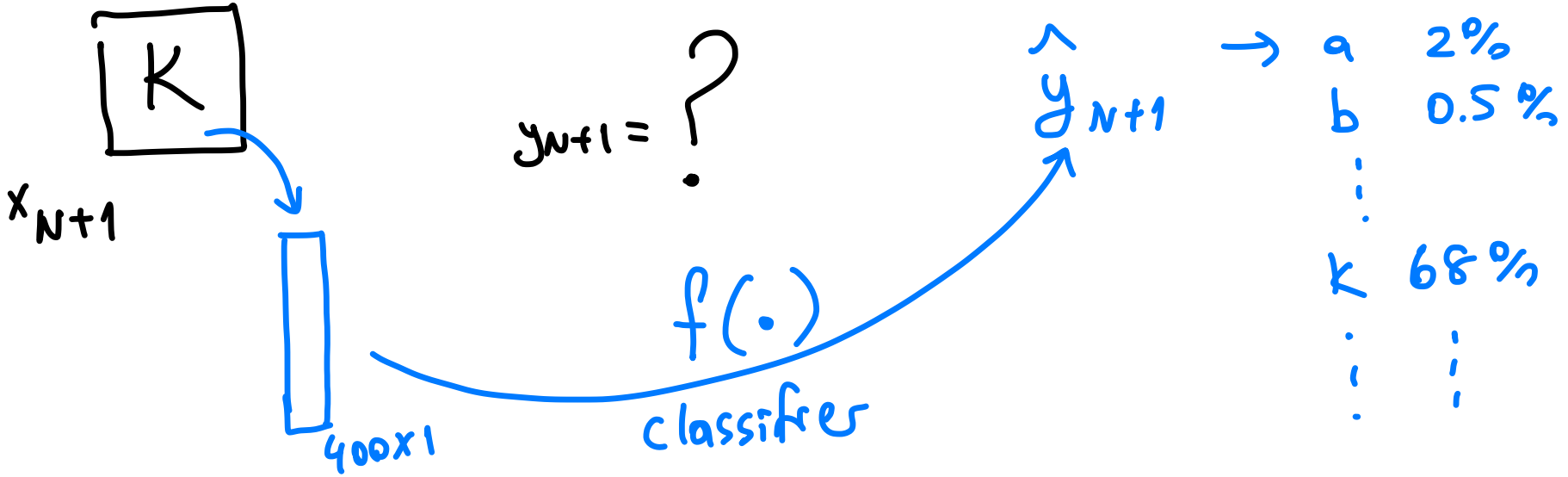
$x_i \in \mathbb{R}^D$

Classification



$x_i \in \mathbb{R}^{400}$ $\forall i$

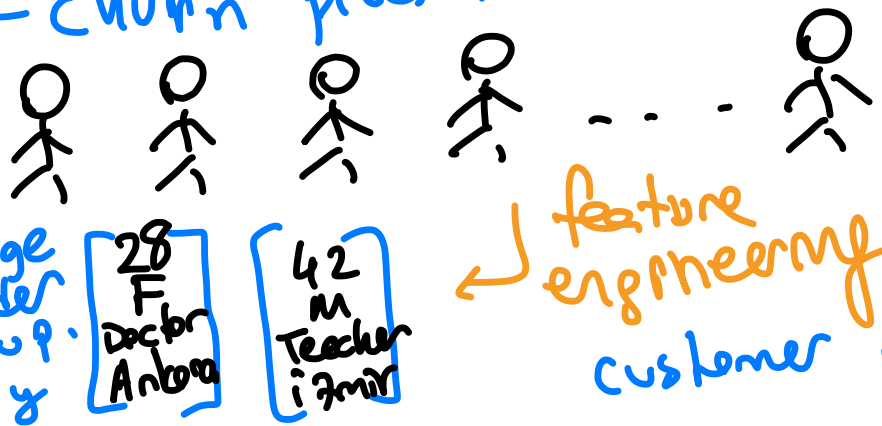
$y_i \in \{a, b, c, \ldots, z\}$



Binary Classification

of classes = 2

- fraud detection
- churn prediction



Multiclass Classification

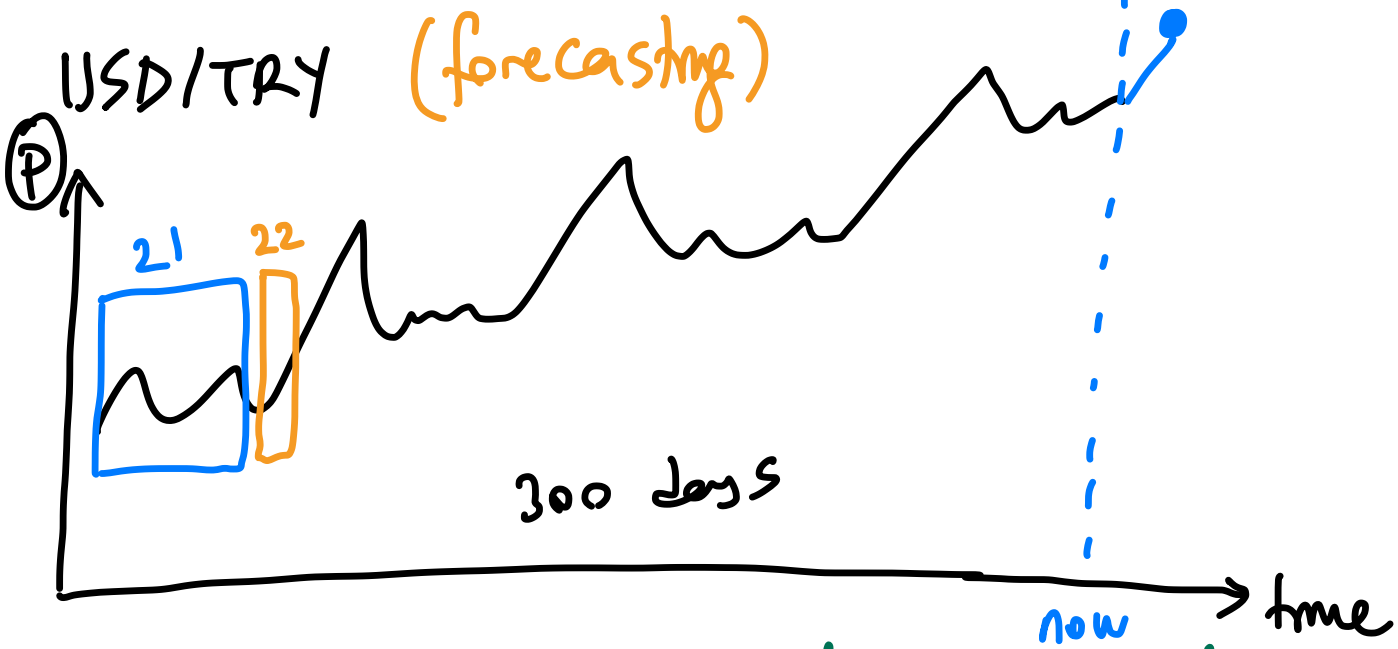
of classes > 2

- digit recognition
- letter recognition
- speech recognition

Regression

$$\mathcal{X} = \{(x_i, y_i)\}_{i=1}^{N}$$

$x_i \in \mathbb{R}^D$ $\quad y_i \in \mathbb{R}$



task: predict USD/TRY value using last 21 days' values

$$x_1 = \begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_{21} \end{bmatrix} \quad x_2 = \begin{bmatrix} p_2 \\ p_3 \\ \vdots \\ p_{22} \end{bmatrix} \quad \cdots \quad x_{279} = \begin{bmatrix} 279 \\ 280 \\ \vdots \\ 299 \end{bmatrix}$$

$$y_1 = [p_{22}] \quad y_2 = [p_{23}] \quad \cdots \quad y_{279} = [p_{300}]$$

$x_i \in \mathbb{R}^{21}$

